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UNIVERSITI TEKNOLOGI MALAYSIA

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UTM Johor Bahru

Semester II 2025/2026

Subject : System Analysis and Design

Section : 01

Task : The FlexRent AI Marketplace

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1.0 Introduction

The Rental System is a web-based application developed to improve the process of booking and managing rental spaces. The system allows users to search, book, and make payments for various types of spaces online, while space owners can efficiently manage listings, pricing, schedules, and availability.

Unlike traditional rental platforms that focus only on a single property type, this system supports multiple categories of rentable spaces such as wedding halls, event venues, meeting rooms, studios, sports facilities, homestays, shop lots, and other rentable spaces.

This project also introduces intelligent features such as dynamic pricing, which automatically adjusts rental prices based on demand, booking periods, and special events. In addition, a smart recommendation system is included to suggest suitable spaces based on user preferences such as budget, location, space type, and ratings. The system also implements a booking schedule and buffer time mechanism to prevent overlapping reservations and ensure proper preparation time between bookings.

Overall, the Rental System aims to provide a more organized, efficient, and user-friendly platform for both customers and space owners.

2.0 Background

Currently, many space rental services still rely on manual methods such as messaging applications, social media platforms, phone calls, or simple spreadsheet records to manage bookings and availability. These methods are often inefficient, difficult to organize, and highly dependent on manual communication between customers and owners.

Customers usually need to contact owners directly to ask about availability, pricing, facilities, and booking details, which can be time-consuming and inconvenient. At the same time, owners may struggle to manage multiple bookings, schedules, and customer requests manually, increasing the possibility of errors and miscommunication.

Furthermore, many existing rental platforms use fixed pricing strategies that do not adapt to demand, peak seasons, holidays, or special events. Most systems also lack personalized recommendations, forcing users to spend more time searching for suitable spaces manually. Without a proper scheduling system, issues such as double bookings, overlapping reservations, and insufficient preparation time may also occur.

Therefore, there is a need for a smarter and more centralized rental management system that can automate booking operations, manage schedules efficiently, improve customer experience, and support multiple types of rentable spaces.

3.0 Problem Statement

The current space rental process faces several issues due to the absence of an integrated and intelligent management system. Firstly, manual booking methods increase the risk of errors such as overlapping reservations, booking conflicts, and communication misunderstandings between customers and space owners.

Secondly, the use of static pricing limits the ability of owners to maximize revenue during high-demand periods or special events. Owners are required to manually adjust prices, which can be inefficient and inconsistent.

In addition, customers often experience difficulty finding suitable spaces because many existing systems do not provide personalized recommendations based on their needs and preferences. This results in longer search times and reduced user satisfaction.

Another major issue is the lack of an efficient scheduling and availability management system. Without automated booking controls and buffer times, spaces may be accidentally booked for overlapping time periods, causing operational problems and reducing service quality.

As a result, both customers and space owners face challenges in terms of efficiency, convenience, booking management, and overall user experience, highlighting the need for a smarter and more automated rental platform.

4.0 Proposed Solution

After careful consideration, the proposed solution for all these troubles is to develop a web-based Rental System, a platform that is designed to ease property booking process for both the customers and providers through automated and AI-integrated systems.

Firstly, the system is implemented with a dynamic pricing system, where it automatically adjusts property prices based on conditions such as peak periods, availability, and demand. This enables the provider to maximise profit while remaining competitive without the hassle of manual updates.

Next, timeslot-based booking is introduced, which allows users to select specific dates and time slots, ensuring accuracy in scheduling stays and preventing double bookings. The system also enforces buffer times between bookings that go hand in hand with the time slot system, ensuring an appropriate time for cleaning and preparations. It is also available with a smart matchmaker system, which suggests the best property for users based on their previous booking behaviour and preferences, such as budget, location, and provided facilities. This system reduces the search time for users, which ensures customer satisfaction.

In addition, the system supports centralised customer and provider management, where all booking details, user profiles, and transaction history are stored and managed efficiently. This prevents miscommunication, reduces errors, and ensures efficient data organization. Lastly, the system provides fast performance and quick loading time, basic data encryption, and a simple interface, ensuring ease of use for the best user experience.

5.0 Objectives

The main objectives of Rental System are:

1. To develop an efficient web-based property booking platform with a structured system that reduces errors.
2. To allow automatic price adjustment based on demands, events, and predetermined conditions which improves revenue management for providers.
3. To provide users with personalized property recommendations.
4. To ensure accurate bookings with timeslot-based booking system and prevent overlapping reservations.
5. To improve booking management for providers and providing tools to manage listings and schedules effectively.
6. To enhance overall user experience.

6.0 Project Scope

- Booking system with time slots
- Instead of just picking a day, users can pick specific times, and it ensures that the house is ready exactly when the guest arrives
- Dynamic pricing
- The system changes prices automatically. This keeps the pricing competitive without the owner having to change it manually every single day
- Smart recommendation system

The system looks at what the user is searching for like "cheap" or "near the city" and highlights homes that fit those needs. It helps guests find what they want faster

- Customer & provider management

It stores profiles for both the guests (customers) and the homeowners (providers). It keeps track of who is staying where, who has paid, and how to contact them if there is a problem.

Not Included:

- Real AI machine learning model

Follows the clear rules set by the programmer. This keeps the project simpler and more reliable for now

- Mobile application

The project is designed to work on a web browser. There won't be a separate app to download from the Apple App Store or Google Play Store

- Integration with external platforms

This project is about building a website for booking Event Spaces, Short-Term Stays, and Long-Term Rentals.

Key features include:

- **Time-slot booking:** Allows guests to pick exact arrival times so the property is ready for them.
- **Smart pricing:** Updates prices automatically based on set rules to stay competitive.
- **Easy search:** Uses simple keywords (like "cheap" or "near the city") to help guests find the right home quickly.
- **User management:** Stores profiles, tracks bookings, and manages contact details for both guests and property owners

7.0 Project Planning

The planning and organization of the project, including the allocation of team members, breakdown of tasks, and overall structure of the development process are carried out. The project exists in several phases, starting from requirement analysis development, testing, and final documentation. Proper planning ensures that the system can be completed within the given timeline and that all team members contribute effectively.

7.1 Human Resources

The project team consists of five members, each assigned with specific roles to ensure smooth development and collaboration.

Member	Role	Responsibilities
Siti Aleeya Qaisara	Project Manager	Manage project progress, gather requirements, ensure system meets objectives
Aimi Nasuha	UI/UX Designer	Design user interface, create prototype, ensure usability
Amny Akalily	Backend Developer	Develop database, implement booking logic, dynamic pricing system
Nurkarmila	Frontend Developer	Develop user interface, integrate frontend with backend
Khairun Najiha Budriah	Tester/Documentation	Conduct system testing, prepare documentation and report

Each member plays an important role in ensuring the successful completion of the project.

7.2 Work Breakdown Structure (WBS)

The project is divided into several major phases, each consisting of specific tasks:

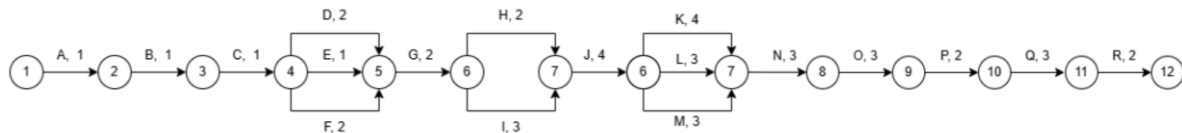
1. Planning Phase
 - Identify project idea
 - Form team and assign roles
 - Set up GitHub repository
2. Requirement Analysis
 - Identify functional and non-functional requirements
 - Define system scope
 - Study existing systems
3. System Design
 - Design system architecture
 - Develop Data Flow Diagram (DFD)
 - Design user interface prototype
4. Development Phase
 - Build user interface
 - Develop booking system with time slots
 - Implement dynamic pricing
 - Implement smart matchmaker
 - Connect system to database
5. Testing Phase
 - Perform functional testing
 - Identify and fix errors
 - Ensure system performance
6. Documentation & Submission
 - Prepare final report
 - Compile project documentation
 - Prepare presentation materials

This structured breakdown helps ensure that all tasks are clearly defined and completed in an organized manner.

7.3 PERT Chart (based on WBS)

The PERT chart is used to illustrate the sequence of activities and dependencies in the project. Each task is assigned an estimated duration and arranged based on its logical order. This helps identify the critical path and ensures that dependent activities are completed in the correct sequence.

Code	Tasks	Predecessor	Duration (days)
A	Identify project idea	-	1
B	Team formation & roles	A	1
C	GitHub repository set up	B	1
D	Identify Requirements	C	2
E	Define system scope	C	1
F	Study existing systems	C	2
G	Design system architecture	D, E, F	2
H	Develop DFD	G	2
I	Design UI prototype	G	3
J	Build UI	H, I	4
K	Booking System	J	4
L	Dynamic pricing	J	3
M	Smart matchmaker	J	3
N	Connect system to database	K, L, M	3
O	Testing	N	3
P	Bug fixes	O	2
Q	Final report & documentation	P	3
R	Presentation preparation	Q	2



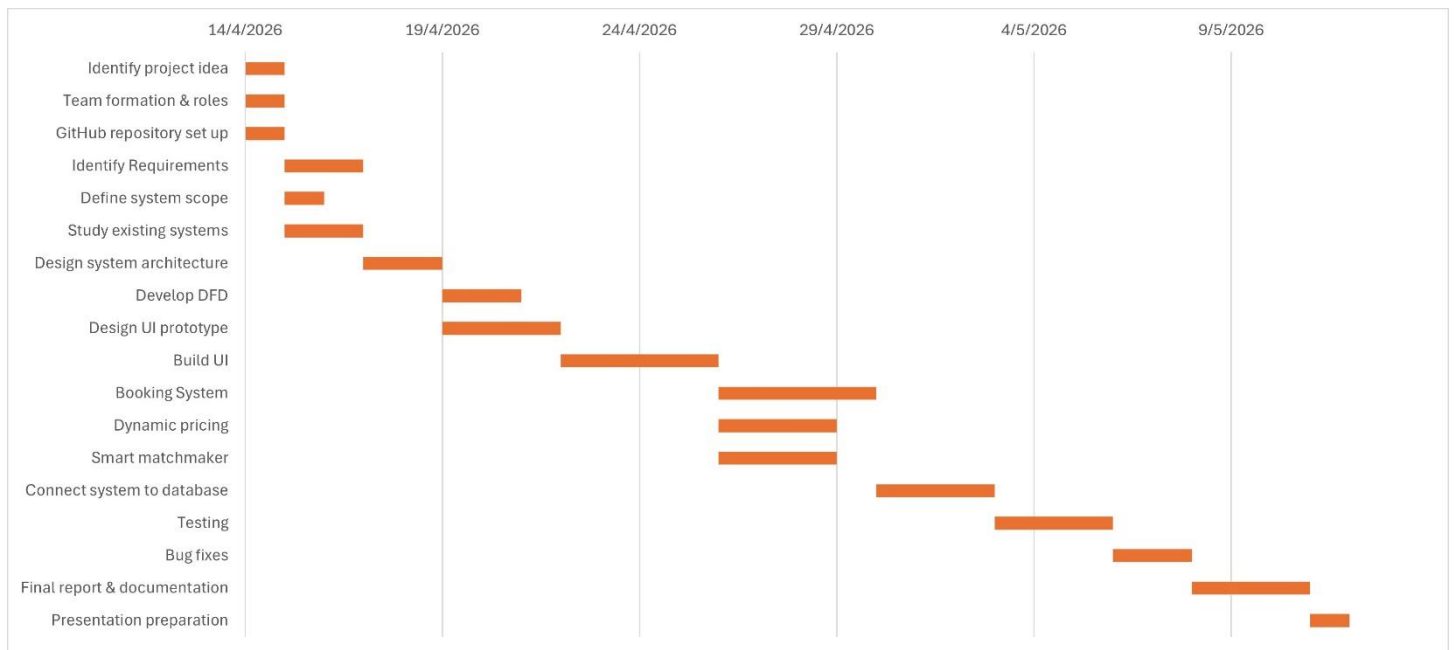
7.4 Gantt Chart

The Gantt chart provides a visual representation of the project schedule, showing the start and end dates of each task. It helps in tracking project progress and ensures that all activities are completed within the planned timeline.

Excel sketch

	A	B	C	D
3	CODE	Tasks	Start Date	Duration (days)
4	A	Identify project idea	14/4/2026	1
5	B	Team formation & roles	14/4/2026	1
6	C	GitHub repository set up	14/4/2026	1
7	D	Identify Requirements	15/4/2026	2
8	E	Define system scope	15/4/2026	1
9	F	Study existing systems	15/4/2026	2
10	G	Design system architecture	17/4/2026	2
11	H	Develop DFD	19/4/2026	2
12	I	Design UI prototype	19/4/2026	3
13	J	Build UI	22/4/2026	4
14	K	Booking System	26/4/2026	4
15	L	Dynamic pricing	26/4/2026	3
16	M	Smart matchmaker	26/4/2026	3
17	N	Connect system to database	30/4/2026	3
18	O	Testing	3/5/2026	3
19	P	Bug fixes	6/5/2026	2
20	Q	Final report & documentation	8/5/2026	3
21	R	Presentation preparation	11/5/2026	1

Gantt Chart



7.5 Project Management Tools

GitHub Repository :

https://github.com/aleeyafayyazz/SmartStay_Project1_SAD_20252026

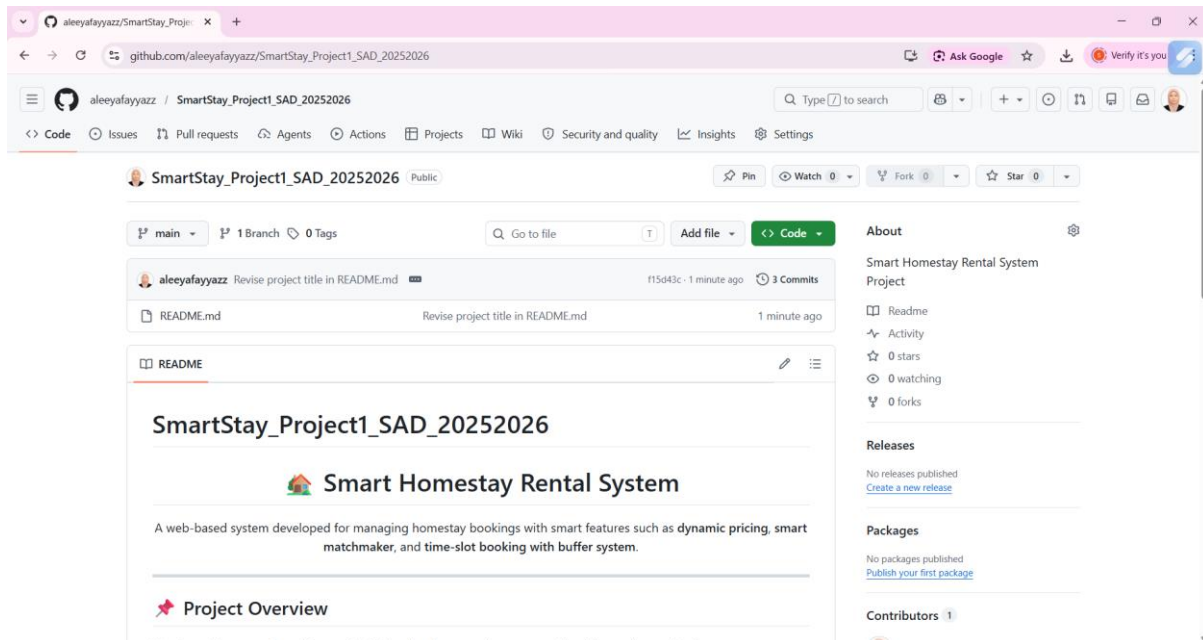


Figure 2.0 : GitHub Repository Interface

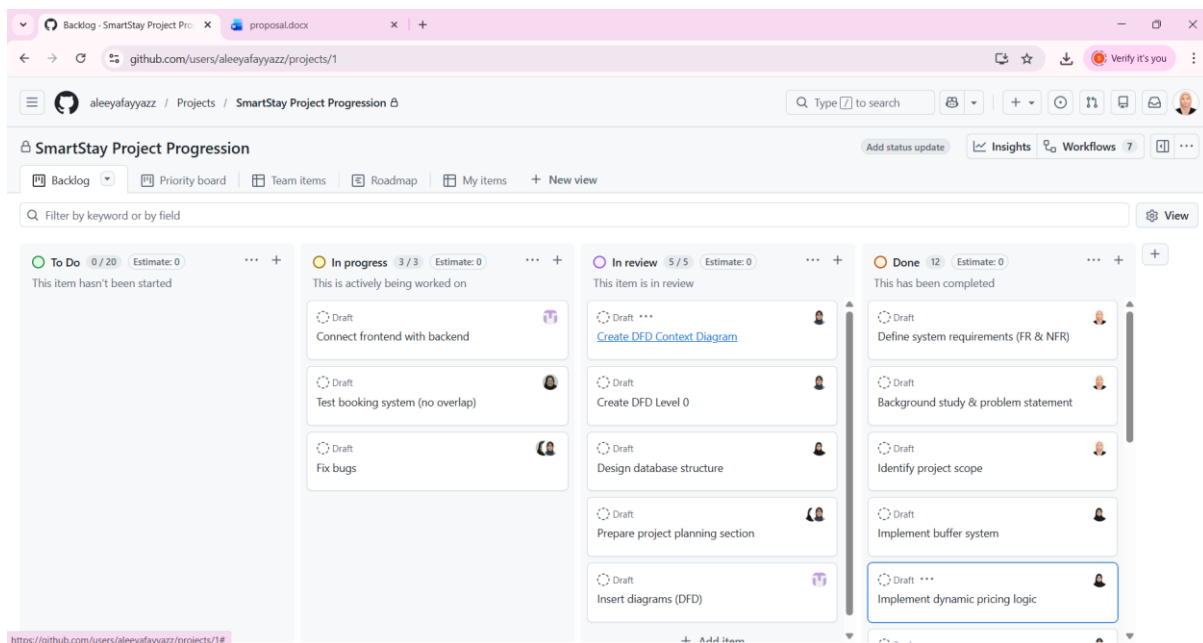


Figure 3.0 : Kanban Board for Task Management

A Kanban board is created in GitHub Projects to manage project tasks. The board consists of columns such as To Do, In Progress, Review, and Done. Tasks are assigned to team members and updated according to the project progress.

8.0 Benefits and Overall Summary of Proposed System

The Smart Homestay Rental System offers several key advantages:

- **Prevents Double Bookings:** The time-slot system acts as a digital gatekeeper. Once a specific time is booked, the system automatically locks it. This prevents accidental double bookings, ensures the property is ready on arrival, and saves hosts from messy scheduling mistakes.
- **Maximizes Profits with Smart Pricing:** Prices update automatically based on set rules. Rates can go up during holidays or busy weekends to match market demand, and drop during quieter weekdays or low seasons to attract more guests.
- **Speeds Up Property Searches:** Instead of scrolling through endless options, guests can type simple keywords like "pool" or "cheap." The system instantly highlights the best matches, making it much faster for users to find what they want.
- **Organizes Data Efficiently:** By keeping all customer profiles, provider details, and payment records in one secure, private place, the system eliminates paperwork and keeps the business running smoothly.

This project is practical, reliable, and highly feasible because it uses proven, rule-based technology rather than complex AI. By focusing on essential web features like automated pricing and exact timeslot scheduling the system directly solves the most common headaches in the rental industry. Guided by a structured project plan and clear team coordination, this platform will deliver a smooth, modern, and stress-free booking experience for both guests and property owners.